

KARTHIK M

Email Id : karthikm270706@gmail.com

Mobile : 9363235005

Virudhunagar, Tamil Nadu

Linked In : www.linkedin.com/in/karthikm027



Carrer Objective :

To work in a dynamic engineering environment where I can utilize my technical knowledge in industrial automation, robotics, PLC systems, and product design to improve operational efficiency, support innovation, and develop myself into a competent engineering professional.

Educational Qualification :

Course	Year of Passing	Institution Name	Board / University	Percentage (%)
B.E	2027 (Pursuing)	Kongu Engineering College	Anna University	8.23 CGPA (upto 6 th semester)
HSC	2023	KCAD Matriculation higher secondary School	State Board	73.83
SSLC	2021	Dr. Radhakrishnan Vidyalyaya Matriculation higher secondary School	State Board	All Pass

Area of Interest:

- PLC & Industrial Automation
- Robotics
- 3D Designing

Technical Skills :

- Siemens S7-1200 PLC
- ABB & FANUC Robot Operation
- Computer Vision System
- Solidworks
- PCB Design and Fabrication

Project Experiences :

- **Title : Machine Vision-Based Intelligent Gripping Force Analysis Using Deep Learning for a Pneumatic Robotic Gripper**
Description : Developed a machine vision-based intelligent gripping system that identifies industrial objects using YOLO and enables predefined gripping-force selection for a pneumatic robotic gripper, improving object handling consistency and operational efficiency.
- **Title : Automated Multi-Stage Mechanical Waste Segregation System**
Description : A multi-stage waste segregation project involves using conveyors and trommel drums to separate biodegradable, non-biodegradable, and metal waste, improving recycling efficiency and supporting sustainable waste management.

➤ **Title : Automated Lemon Grading and Size-Based Segregation System**

Description : Developed an automated grading system that sorts lemons into multiple size categories using a rotating trommel, ultrasonic sensors, and embedded control. The solution increases grading consistency, minimizes sorting errors, and improves productivity for agricultural and export applications.

➤ **Title : Title: Automated Fabric Defect Inspection and Marking System**

Description: Developed an automated fabric inspection and defect-marking system for detecting holes, oil spots, stains, and surface defects. Implemented controlled lighting and image-based computational analysis to identify and locate defects, followed by an automated stamping mechanism to precisely mark the defective regions.

Mini Projects:

➤ **Title: Digital door lock**

Description: A smart security system developed using an 8051 microcontroller for password-based door access through a keypad, with motor-controlled locking and status display on an LCD screen.

➤ **Title: Room Safety Monitoring System**

Description: An Arduino-based system that monitors gas, fire, temperature and humidity, providing real-time alerts and remote monitoring through Wifi.

Achievements:

- Selected as a Finalist in **IEEE YESIST12 2026 – IEngage Track** (IEngage Direct Entry) for the project “Smart IoT-Enabled Multi-Stage Waste Segregation and Plastic Cleaning System for Sustainable Community Waste Management”, selected among 2,000+ abstract submissions.
- Won First Prize in Badminton “**INTRAMURAL TOURNAMENTS**” conducted by Kongu Engineering College.
- Won First Prize in Paper Presentation in Intra College Technical Symposium “**FRISSION 2K25**” Conducted by Food Technology Association at Kongu Engineering College.
- Won First Prize in Project Presentation in Intra College Technical Symposium “**FRISSION 2K25**” Conducted by Food Technology Association at Kongu Engineering College.
- Won First Prize in Badminton “**INTRAMURALTOURNAMENTS**” conducted by Kongu Engineering College.

Co-Curricular Activities:

- Participated in “**IDEATHON ‘24**” organized by the Mechatronics Engineering Association of Kongu Engineering College.
- Participated in a Tech Talk event during “**THROB24**” an Intra-Departmental Symposium organized by the Mechatronics Engineering Association of Kongu Engineering College.

Workshop:

- Participated in a Two days workshop on “**MOBILE ROBOT**”, gaining hands-on experience in mobile robot operation, motion control, and practical robot handling.
- Participated in the Two days workshop on “**THE ROBO SUMO**” organized by the Robotics Club of Kongu Engineering College.
- Participated in the workshop on “**ELECTROCAD**” organized by the Mechatronics Engineering Association of Kongu Engineering College.

Certification:

- Successfully completed the **Certified SolidWorks Associate (CSWA)** international certification.

Interpersonal Skills :

- Team Leadership
- Quick Learner
- Punctuality